

Galactic Structure, Galaxies & the ISM

Problem Set — Solutions

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Section A — Objective

O1. Answer: (a). In seconds, $(t_4 - t_1) = 21600$ s, $(t_3 - t_2) = 7200$ s:

$$R_l = \frac{v}{4} [(t_4 - t_1) + (t_3 - t_2)] = \frac{50}{4} (21600 + 7200) \approx 3.6 \times 10^5 \text{ km.}$$

O2. Answer: (b). Distance alone dims a star without reddening it; only dust preferentially scatters blue light, producing both effects together.

O3. Answer: (a). $L_{\text{Edd}} = 4\pi GMc/\kappa$ with $M = 10M_\odot = 1.989 \times 10^{31}$ kg:

$$L_{\text{Edd}} = \frac{4\pi(6.674 \times 10^{-11})(1.989 \times 10^{31})(3 \times 10^8)}{0.034} \approx 1.5 \times 10^{32} \text{ W.}$$

O4. Answer: (d) is false. Seyferts, quasars, and blazars share the *same* central engine — an accreting supermassive black hole near its Eddington limit — differing only in luminosity, host prominence, and viewing angle. (a), (b), (c) are all correct.

O5. Answer: (a). $M(r) = \Theta^2 r / G$ with $\Theta = 2.2 \times 10^5$ m/s, $r = 15$ kpc $= 4.629 \times 10^{20}$ m:

$$M(r) = \frac{(2.2 \times 10^5)^2 (4.629 \times 10^{20})}{6.674 \times 10^{-11}} \approx 3.36 \times 10^{41} \text{ kg} \approx 1.7 \times 10^{11} M_\odot.$$

Section B — Subjective

S1.

(i) $M_1/M_2 = K_2/K_1 = 120/80 = 1.5$.

(ii) In SI units, $f(M_1) = PK_1^3/(2\pi G) = (4.32 \times 10^5)(8 \times 10^4)^3/(2\pi \times 6.674 \times 10^{-11}) \approx 5.28 \times 10^{29}$ kg $\approx 0.265 M_\odot$. With $M_1 = 1.5M_2$, so $M_1 + M_2 = 2.5M_2$:

$$\frac{M_2^3}{(2.5M_2)^2} = \frac{M_2}{6.25} = 0.265 \implies M_2 \approx 1.66 M_\odot, \quad M_1 \approx 2.49 M_\odot.$$

- (iii) $M_1 + M_2 \approx 4.15 M_\odot$; with $P = 0.0137$ yr, Kepler's Third Law gives $a^3 = (4.15)(0.0137)^2 \approx 7.77 \times 10^{-4} \text{ AU}^3 \implies a \approx 0.0918 \text{ AU} \approx 1.37 \times 10^7 \text{ km}$. Then

$$v = \frac{2\pi a}{P} \approx 200 \text{ km/s} = K_1 + K_2 \checkmark.$$

(This must hold at $i = 90^\circ$: each star's own radial velocity amplitude *is* its full orbital speed at conjunction, so the total relative speed is just the sum, $K_1 + K_2 = 2\pi(a_1 + a_2)/P = 2\pi a/P$.) Using $(t_4 - t_1) = 36000 \text{ s}$, $(t_3 - t_2) = 10800 \text{ s}$:

$$R_l = \frac{200000}{4}(36000 + 10800) \approx 2.34 \times 10^9 \text{ m} \approx 3.4 R_\odot,$$

$$R_s = \frac{200000}{4}(36000 - 10800) \approx 1.26 \times 10^9 \text{ m} \approx 1.8 R_\odot.$$

S2.

- (i) With $(d\Theta/dR)_0 = 0$: $A = \Theta_0/(2R_0)$, $B = -\Theta_0/(2R_0)$, so $A + B = 0$ identically.
- (ii) $A + B = 15 + (-13) = 2 \text{ km/s/kpc} = -(d\Theta/dR)_0 \implies (d\Theta/dR)_0 \approx -2 \text{ km/s/kpc}$. This is close to, but not exactly, zero — the Milky Way's rotation curve near the Sun is very nearly flat, with only a slight residual decline.
- (iii) Writing both out: $A = \frac{1}{2}(\Theta_0/R_0) - \frac{1}{2}(d\Theta/dR)_0$ and $B = -\frac{1}{2}(\Theta_0/R_0) - \frac{1}{2}(d\Theta/dR)_0$ — notice *both* carry the identical term $-\frac{1}{2}(d\Theta/dR)_0$. Subtracting cancels this shared term exactly, leaving pure Θ_0/R_0 ; adding instead would double it, which is exactly why $A + B$ isolates the shear term instead.

S3.

- (i) $M = L\kappa/(4\pi Gc) = (10^{39})(0.034)/[4\pi(6.674 \times 10^{-11})(3 \times 10^8)] \approx 1.35 \times 10^{38} \text{ kg} \approx 6.8 \times 10^7 M_\odot$.
- (ii) $\dot{M} = L/(\eta c^2) = 10^{39}/[0.1(9 \times 10^{16})] \approx 1.11 \times 10^{23} \text{ kg/s} \approx 1.8 M_\odot/\text{yr}$.
- (iii) $R_s = 2GM/c^2 = 2(6.674 \times 10^{-11})(1.35 \times 10^{38})/(9 \times 10^{16}) \approx 2.0 \times 10^{11} \text{ m} \approx 1.3 \text{ AU}$. Even a $\sim 7 \times 10^7 M_\odot$ black hole — among the most massive objects in the universe — has an event horizon only about the size of Earth's own orbit, a striking reminder of just how extreme the compactness of these objects is.